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Hyperspectral Anomaly Detection Based on Orthogonal Subspace Projection with Compressive Sampling

Xiao Zhang, Meiping Song and Haimo Bao

***Abstract*—The low-rankness and sparsity are the typical characteristics of hyperspectral image, corresponding to background pixels and distinctive features pixels such as small targets and noise. However, the effective separation of these three components remains a challenging problem. To address these challenges, we propose a novel hyperspectral anomaly detection method based on the orthogonal subspace projection with compressed sampling (RSOSP). The RSOSP utilizes projection strategies to generate the compressed sensing band domain (CSBD), in which the background pixels are further solved in CSBD and sparse component in the original observation space is obtained via orthogonal subspace projection. In addition, RSOSP extracts the remaining sparse component in CSBD using the orthogonal complement of the subspace spanned by the background component, and further selects the effective anomaly space via hard-thresholding processing. Based on the pixels existing in CSBD, orthogonal subspace and anomalies space, a detection method is designed using Mahalanobis distance between the observed pixel and the background pixels, which involves multiple detectors. Meanwhile, we continuously stack the obtained anomaly detection maps onto the original HSI image, thereby enhancing the representability of anomaly target. Finally, RSOSP attains a satisfactory or expected detection performance with little computational cost. Experimental results on six real hyperspectral datasets verify that the proposed method is more effective than the comparison methods.**

***Index Terms*—Hyperspectral anomaly detection (HAD), decomposition, compressed sampling, subspace projection.**

I. INTRODUCTION

DIFFERENT from other remote sensing data obtained by various Earth observation technologies, hyperspectral imagery (HSI) exhibits two-dimensional spatial distribution information and abundant spectral reflectance characteristics of the ground-objects [1], [2]. These make it possible to accurately identify and distinguish objects [3]. Hyperspectral anomaly detection (HAD) aims to effectively identify anomaly targets that are usually different from background pixels and occur with low probability[4], [5]. Various HAD methods have been developed[6] including deep learning-based methods [7], [8], [9], statistical-based methods[9], [11], representation-based methods[12], [13], decomposition-based methods[14], [15], and subspace-based methods[16], [17].

In recent years, deep learning-based methods have shown obvious advantages. Due to the lack of labeled samples, unsupervised and self-supervised methods have become increasingly dominant [18], [19]. Among them, autoencoders-based (AEs) [20], [21] and generative adversarial networks-based (GANs) [22], [23] methods have been widely applied. AE-based methods usually learn background representations through reconstruction and identify anomalies based on large reconstruction errors. For instance, Tu et al. [24] proposed a self-supervised graph attention autoencoder to detect anomalies via reconstruction deviations. Some recent studies further enhance background reconstruction to better suppress anomalies. Fu et al. [25] proposed an unsupervised multiscale Mamba reconstruction network that enhances background representation through random masking, multiscale background enhancement, and dynamic feature fusion. For GAN-based methods, Wang et al. [26] adopted an unsupervised GAN to model background distribution via adversarial learning and further introduced continual learning for cross-domain anomaly detection. In addition, some studies combine deep learning with traditional models. Fu et al. [27] proposed an unsupervised hyperspectral anomaly detection method that embedded a CNN denoiser into a traditional low-rank representation model. Emoto and Matsuoka [28] integrated tensor robust principal component analysis (TRPCA) with autoencoding adversarial networks and weighted local RX for unsupervised anomaly detection. Recently, spatial context information has also attracted increasing attention [29]. For example, Wang et al. [30] developed a self-supervised blind-spot network that combines local spatial–spectral features with nonlocal long-range dependencies. Deep-learning-based methods have improved the anomaly detection performance, while parameter adjustment remains time-consuming. The statistical anomaly detection methods assume that background pixels follow a Gaussian distribution [31], [32]. A series of improved Reed-Xiaoli (RX)-based methods [33], local RX (LRX)-based methods [34], kernel RX (KRX)-based methods [35], and weighted RX (WRX)-based methods [36] have been

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developed. However, in most real HSI scenes, the statistical distribution cannot accurately describe the complex background, leading to unsatisfactory detection performance.

Sparse representation (SR)-based methods, constraint representation (CR)-based methods and low-rank representation (LRR) methods measure anomalies via residuals between the reconstructed pixel and the original pixel. SR-based methods apply as few background dictionary atoms as possible to represent background pixels, while anomalies cannot be represented in this way [37]. For instance, Wang et al. [38] leveraged quaternion-guided sparse representation to encode both spatial priors and super-band spectral structures, thereby improving anomaly–background separability via structured sparse recovery. To make sufficient use of the spatial information, Ling et al. [39] modeled each background pixel as a sparse linear combination of its spatial neighbors and detected anomalies via reconstruction errors after adaptively purifying the background dictionary. In [40], joint sparse representation enforces a common sparsity pattern within homogeneous regions for adaptive background modeling and anomaly detection. Li and Du [41] proposed a classic CR detector (CRD), which adopts a double window strategy to detect anomalies. CR-based methods express background pixel with the help of neighboring pixels, representing each background pixels using all samples in the background dictionary [42], [43]. For example, Hou et al. [44] combined the CRD with background purification and the saliency weight method, which adopts the classical least squares technique to obtain the preliminary linear representation coefficients and purify the background. Without the consideration of the global correlation of all pixels in HSI data, the detection results of these methods may have a high false alarm rate when the constructed background dictionary is corrupted by anomalies. The LRR-related models realize a lowest-rank representation of all HSI pixels, with the high spatial and spectral correlations of HSI. Zhang et al. [45] introduced the low-rank dictionary to reconstruct the background pixels rather than anomaly pixels. Xie et al. [46] proposed a dictionary-based LRR, which contains observed training data and hidden learned data drawn by the background estimation model. LRR-related methods apply the global data characteristics to represent HSI from multiple subspaces[47].

Furthermore, the LRR models are transformed into low-rank and sparse matrix decomposition ones. Considering that the background pixels and anomalies in HSI are low-rank and sparse, respectively, the robust principal component analysis (RPCA) algorithm decomposes HSI data into the low-rank item and sparse item[48]. The low-rank and sparse matrix decomposition (LRaSMD) method further separates the noise from the sparse component [49], [50]. Different from the LRaSMD method, the LSMAD method adopts the Mahalanobis distance to detect anomalies[51]. Su et al. [52] proposed a relaxed collaboration representation (RCR) method, which utilizes LRaSMD idea to separate the anomalies and background in dictionary construction. Wang et al. [53] constrained LRaSMD with reweighted nuclear norm and total variation regularizations. Some low-rank and sparse decomposition methods can also accurately detect anomalies via the mixture-of-Gaussian distribution (MoG)[54], [55], [56]. Specifically, MoG could model the sparse residual component into mixture of several Gaussian components, and select the most likely anomaly component through standard deviation or contrast. However, the obtained sparse item may cover some extra background pixels that exhibit sparse characteristics. Incomplete separation often exists in this type of method.

In recent years, many subspace-based methods have been developed for anomaly detection, including tensor subspace methods, dictionary representation-based subspace methods, orthogonal subspace projection (OSP)-based methods, and other subspace methods [57], [58], [59]. First, tensor subspace methods preserve the spatial-spectral structure and utilize multiple subspaces. Ma et al. [60] realized the decomposition of the background tensor and anomaly tensor via the tensor subspace learning method. By combining the dictionary representation, Guo et al. [61] projected the prior background into a low-rank subspace which is constructed by background endmembers and detected anomalies via residuals. By adopting OSP technologies, Chang et al. [62] took go decomposition under the orthogonal subspace, and proposed an OSP-version go decomposition method. Yang et al. [63] proposed a LRaSMD with OSP-based background suppression and adaptive weighting for HAD, which projects the sparse component into the background subspace, contributing to background suppression and anomaly detection. Zhang et al. [64] constructed three background subspaces and obtained their orthogonal subspaces, thus combining three projection results into the final AD scores. Thus, these methods achieve effective separation between anomalies and background in the adequate subspace, which not only detect anomalies more accurately, but also avoid the unnecessary interference, for example, the boundaries and noise. In addition, the band selection constructs the subspace with a better separation performance than the original band sets. In [65], Shang et al. proposed a residual driven BS (RDBS) method, which obtains a band subset with high representation capability to improve the anomaly detection. In [66], Zhang et al. constructed an out-of-distribution based feature subspace representation module to acquire latent feature spaces with a better indication of the anomaly detection ability. Similarly, the feature projection realizes the projection from the original data space to a new feature space, thus achieving the separation of the background, anomalies and anomaly target pixels. In [67], Ma et al. projected the original data into a low-dimensional feature space to mitigate spectral redundancy, thereby realizing the low-rank representation and better separating anomalies from the background. In [68], Li et al. integrated the background dictionary learning into HAD, which optimized the background dictionary via the projection operators and overcome the limitations of statistical dictionaries in traditional methods. In [69], Li et al. projected the high-dimensional background spectrum onto a low dimensional subspace by learning a low-dimensional orthogonal basis V,

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where the background was represented by subspace coefficients U. As mentioned above, the band selection and feature projection-based subspace methods promote the separation of background and anomalies by constructing low dimensional subspaces. However, these generated subspaces may contain heterogeneous interference and mixed spectral information, which weaken the discriminative ability of subspace and prevent sufficient decomposition of the low-rank background components and sparse anomaly components.

In this article, to select significant and reasonable pixels as the background component, we adopt compressed sampling to construct the background model based on the low-rank property of background pixels. Besides, considering that the separation of low-rank component and sparse component is still a problem that needs to be solved in HAD tasks, we further project the obtained background pixels into the orthogonal subspace to enhance the separation. The feature projection-based subspace representation and OSP both contribute to increasing the differentiation of background pixels and anomalies to a certain extent. Thus, we proposed a hyperspectral anomaly detection method based on OSP with random sampling processing (RSOSP). Inspired by the LRaSMD idea, RSOSP achieves the solution of the low-rank component and sparse component of HSI data, and constructs a model containing multiple detectors based on the Mahalanobis distance. Due to the fact that the subspace decomposition improves the separation of background pixels and anomalies, the background suppressibility and target detection abilities can be further enhanced. Last but not least, RSOSP gradually extracts the corresponding anomaly information to enrich the original HSI data, thus optimizing the final detection result. The main contribution of the proposed RSOSP can be summarized as follows:

1) The proposed method adopts compressed sampling data to characterize the background structure of hyperspectral image because the sampled data preserves the dominant structure and main body information of the original HSI. To further obtain a purer background component, RSOSP imposes a low-rank constraint on the compressed sampling data. Meanwhile, the reduced dimensionality of the data reduces the subsequent computational cost.
2) RSOSP directly extracts complementary sparse components through orthogonal projecting in dual-domains. The major sparse component is obtained in the original HSI data space by projecting the data onto the orthogonal complement of the subspace defined by the approximate background representation, while the remaining sparse component is further extracted in the compressed sensing band domain based on the orthogonal complement of the subspace spanned by the refined background component. Subsequently, we separate the subtle sparse items with the help of the hard-thresholding processing. This enhances the separability between background pixels and sparse anomalies as much as possible.
3) For detecting anomalies, we construct multiple detectors with the generated decomposition result, and take the generated detection result map as the new band of the pending HSI data. Multiple feedback operations enhance the input data by exploiting background and anomaly response information in the generated anomaly detection map, which can provide supplementary discriminative information when the current detection result can still be improved and the feedback information is reliable.

The remaining part of this paper is organized as follows: Section II introduces the related methods. Section III describes the proposed algorithm. Then, Section IV presents the experimental analysis. Section V concludes this paper.

II. Related Method

A. Compressed sensing sampling

The compressed sensing (CS) realizes the direct collection of the HSI data, and reduces the burden of the high sampling rate in many applications [70]. The CS indicates that, when a signal can be sparsely represented in a certain transformation domain, the signal sampled by the observation matrix that satisfies the restricted isometry property (RIP) can represent most of the information of the original signal.

Specifically, the signal can be sampled by the observation matrix when the signal is sparse. If the signal $x \in \mathrm{R}^N$ to be compressed may not be sparse or compressible, it is necessary to search for a sparse basis $\psi \in R^{N\times N}$ to make x sparse in ψ domain, as follows:

$$x = \sum_{i=1}^{N} \psi_i \alpha_i = \psi\alpha \tag{1}$$

where $x = [x_1, x_2, ..., x_N]$, $\psi = [\psi_1, \psi_2, ..., \psi_N]$, and α are the sparse coefficient matrix. In the sparse basis ψ, x is sparse when there are only M nonzero item in α, where *M* is much smaller than *N*. The sparse representation of the signal can be realized by an orthogonal basis dictionary and an overcomplete dictionary.

Next, the compressed sampling utilizes the observation matrix to collect the important information of the signal, performs dimensionality reduction processing, and obtains the compressed measurement vector $y \in R^M$ as follows:

$$y = \phi x \tag{2}$$

where the observation matrix ϕ is usually uncorrelated to the sparse basis ψ. And (2) can be written as (3), where $A \in R^{M\times N}$ is the sensing matrix.

$$y = \phi\psi\alpha = A\alpha \tag{3}$$

When the signal is not sparse, the sparse basis projects the original signal into sparse domain. Thus, the observation matrix needs to meet the uncorrelation condition of the observation matrix and sparse basis. Meanwhile, the observation matrix needs to have the RIP as follows:

$$(1-\delta_\rho)\|x\|_2^2 \ll \|\phi x\|_2^2 \ll (1+\delta_\rho)\|x\|_2^2 \tag{4}$$

When equidistant coefficient $\delta_\rho \in (0,1)$, $\rho = 1, 2, ..., \mathrm{P}$, it is considered to satisfy the ρ -order limited equivalence condition.

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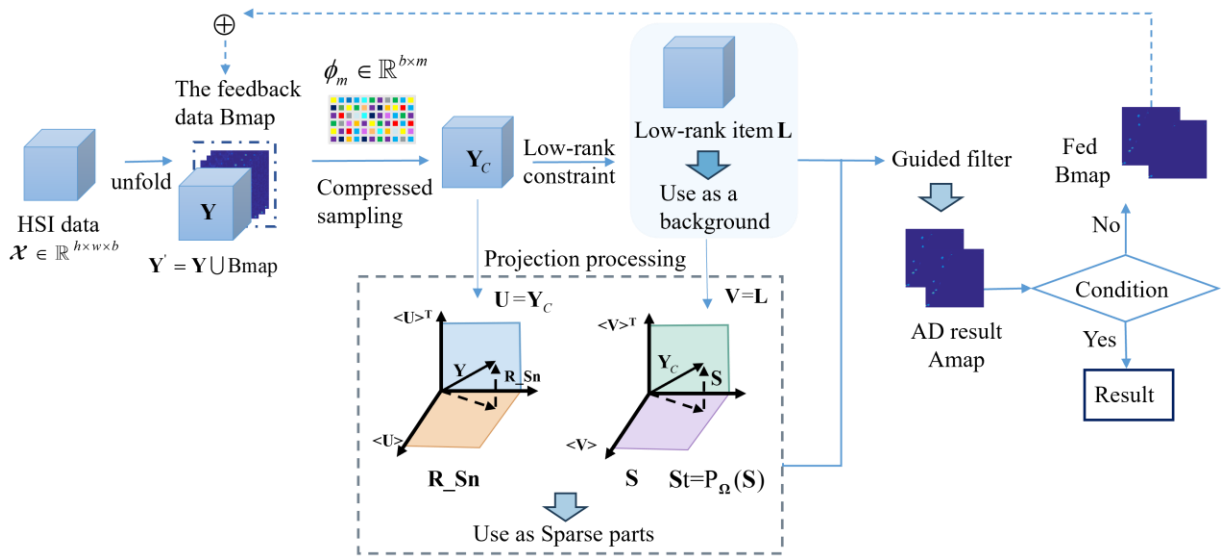


Fig.1. The framework image of RSOSP.

The effective observation matrix wouldn't destroy the original structure of the signal. The less correlated the observation matrix is with the sparse basis, the higher the reconstruction accuracy in the following processing. As one of the random observation matrixes, the Gaussian random matrix obeys the independent normal distribution, which is not related to most sparse basis. Besides, the Bernoulli's random matrix and binary random matrix are similarly selected as the observation matrix in CS processing.

Last, the signal reconstruction utilizes the optimization algorithm to solve the original signal using some compressed measurement value, shown as follows:

$$\begin{aligned} &\min \|x\|_0 \\ &s.t.\ y = \phi x \end{aligned} \tag{5}$$

However, the solution of (5) is an NP-hard problem. Candes et al proves that (5) is equal to (6) when the observation matrix satisfies RIP [71]:

$$\begin{aligned} &\min \|x\|_1 \\ &s.t.\ y = \phi x \end{aligned} \tag{6}$$

Thus, the compressed measurement vector y can accurately restore signals.

According to the CS theory, much fewer compressed measurement value than the sampled value obtained by the Nyquist-Shannon sapling theorem can probably reconstruct the signal, when the signal is sparse or show sparsity in certain transform domain.

As mentioned above, the CS technology mainly contains the following three steps: the sparse expression of the original signal, the design of the observation matrix, and the reconstruction of the signal. The first two steps obtain the compressed measurement value which realize the preservation of the critical information in a lightweight size. Besides, considering that the HSI data also can be sparsely represented by a linear combination of a certain dictionary, the compressed sampling is also suitable for HSI imaging. Based on this, Chang et al. simplified CS method (SCS) for HSI processing by only applying signal sampling technology as follows [72]:

$$y = \phi_R x + n \tag{7}$$

where ϕ_R is a random sampling matrix.

B. Orthogonal subspace projection technology

In real hyperspectral scenes, the pixels y are usually not pure substance; they could be represented by a linear combination, as shown in (8),

$$y = \alpha_1 m_1 + \alpha_2 m_2 + ... + \alpha_p m_p + \mathrm{n} \tag{8}$$

where $m_1, m_2, ..., m_p$ denotes the unrelated linear spectral vectors with R bands, and $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_p$ is the corresponding sparse coefficient vector, n represents noise. Harsanyi and Chang proposed the orthogonal subspace projection (OSP) operator, which realizes the removal of the undesired substance in y [73]:

$$P_{\mathbf{U}}^{\perp} = 1 - \mathbf{U}\mathbf{U}^{\#} = \mathrm{I} - \mathbf{U}\left(\mathbf{U}^{\mathrm{T}}\mathbf{U}\right)^{-1}\mathbf{U}^{\mathrm{T}} \tag{9}$$

where $P_{\mathbf{U}}^{\perp}$ is the OSP operator, $\mathbf{U} = \left[m_1, m_2, ..., m_{p-1}\right]$ represents the uninterested substances, and $\mathbf{U}^{\#}$ is the pseudo inverse of $\mathbf{U}$. Thus, y can be projected onto the orthogonal subspace $\langle \mathbf{U} \rangle^{\mathrm{T}}$ by the OSP operator $P_{\mathbf{U}}^{\perp}$.

III. Proposed Method

Traditional low-rank and sparse methods are realized by RPCA, GoDec or the sparse expression. The generated low-rank component and sparse component are constrained by the low-rank prior and different norms. Given that the compressed sensing data exhibit low-rank characteristics, we further take orthogonal subspace projection processing. The projected results thereby retain the structural characteristics analogous to the sparse and noisy terms in low rank sparse matrix factorization. Thus, this paper proposes a random compressed sampling based orthogonal subspace projection anomaly detection method (RSOSP).

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A. RSOSP

For a hyperspectral image $\mathcal{X} \in \mathbb{R}^{h\times w\times b}$, its spatial size is $h\times w$ with b spectral bands. It can be unfolded into a two-dimensional matrix $\mathbf{Y} = unfold(\mathcal{X}) \in \mathbb{R}^{D\times b}$, where $D = h\times w$ denotes the total number of pixels. The RSOSP aims at decomposing the HSI $\mathbf{Y}$ under different subspaces, thus generating low-rank component and multiple sparse components.

Due to the fact that $\mathbf{Y}$ could be linearly represented by different substances, we assume that HSI data $\mathbf{Y}$ can be linearly represented by $m_1, m_2, ..., m_{p-i}$ unrelated background spectral vectors and $m_{p-i+1}, m_{p-i+2}, ..., m_p$ uncorrelated anomaly spectral vectors.

$$\begin{aligned}\mathbf{Y} &= \mathbf{M}\alpha + \mathrm{n} \\ &= \left(\alpha_1 m_1 + \alpha_2 m_2 + ... + \alpha_{p-i} m_{p-i}\right) + \\ &\left(\alpha_{p-i+1} m_{p-i+1} + \alpha_{p-i+2} m_{p-i+2} + ... + \alpha_p m_p\right) + \mathrm{n}\end{aligned} \tag{10}$$

From (10), we can infer that $\mathbf{Y}$ could be sparsely represented. Thus, we adopt the compressed sensing technology to obtain the approximation expression of $\mathbf{Y}$ with a low dimension.

Under the RIP condition, we design the Gaussian sampling matrix $\phi_m \in \mathbb{R}^{b\times m}$ to compress the HSI data $\mathbf{Y}$ with the random sampling matrix. In the compressed sensing band domains (CSBDs), we regard the compressed data of $\mathbf{Y}$ as $\mathbf{Y}_{CSBDs} \in \mathbb{R}^{D\times m}$. Thus, the compressed observation in CSBDs can be solved by the following compressed sampling:

$$\mathbf{Y}_{CSBDs} = \mathbf{Y}\phi_m \tag{11}$$

The value of m can be approximately defined by Harsanyi-Farrand-Chang (HFC) or noise-whitened HFC (NWHFC) methods according to the VD theory [74], [75].

Then, the background component $\mathbf{L} \in \mathbb{R}^{D\times m}$ is solved by the low-rank constraint of $\mathbf{Y}_{CSBDs}$ as follows:

$$\mathbf{L} = \arg\min_{\mathbf{L}} \left\{ \left\| \mathbf{Y}_{CSBDs} - \mathbf{L} \right\|_F^2 + rank(\mathbf{L}) \right\} \tag{12}$$

To obtain the separation of the sparse component, we further combine subspace projection strategy to solve the remaining information in the original HSI data domain except $\mathbf{Y}_{CSBDs}$. The projection processing is conducted through the following projection matrix $\mathrm{P}_{\mathbf{Y}_{CSBDs}}^{\perp}$:

$$\mathrm{P}_{\mathbf{Y}_{CSBDs}}^{\perp} = \mathbf{I} - \mathbf{Y}_{CSBDs}\left(\mathbf{Y}_{CSBDs}{}^{\mathrm{T}}\mathbf{Y}_{CSBDs}\right)^{\dagger}\mathbf{Y}_{CSBDs}{}^{\mathrm{T}} \tag{13}$$

in which $(\cdot)^{\dagger}$ represents pseudo inverse.

Thus, sparse component $\mathbf{R_Sn} \in \mathbb{R}^{D\times b}$ can be obtained through projection as follows:

$$\mathbf{R_Sn} = \mathrm{P}_{\mathbf{Y}_{CSBDs}}^{\perp} \cdot \mathbf{Y} \tag{14}$$

Here, $\mathbf{R_Sn}$ is the residual component of $\mathbf{Y}$ outside the estimated background subspace spanned by $\mathbf{Y}_{CSBDs}$.

On the other side, $\mathbf{Y}_{CSBDs}$ is generated from compressed sampling, which may be not correspond to pure background item. We assume that $\mathbf{Y}_{CSBDs}$ is composed of background approximation $\mathbf{L}$, anomalies, noise and other disturbances, shown as follow:

$$\mathbf{Y}_{CSBDs} = \mathbf{L} + \mathbf{E}_c \tag{15}$$

in which $\mathbf{E}_c$ denotes anomalies, noise and disturbance. Then, we obtain the second sparse component $\mathbf{S} \in \mathbb{R}^{D\times m}$ as the projection result of $\mathbf{Y}_{CSBDs}$ under the projection matrix $\mathbf{I} - \mathrm{P}_{\mathbf{L}}$:

$$\mathbf{S} = (\mathbf{I} - \mathrm{P}_{\mathbf{L}})\mathbf{Y}_{CSBDs} + \varepsilon_{\mathbf{L}} \tag{16}$$

$$\mathrm{P}_{\mathbf{L}} = \mathbf{L}\left(\mathbf{L}^{\mathrm{T}}\mathbf{L}\right)^{\dagger}\mathbf{L}^{\mathrm{T}} \tag{17}$$

Here, $\mathrm{P}_{\mathbf{L}}$ denotes the estimated projection operator from $\mathbf{L}$, $\varepsilon_{\mathbf{L}}$ represents the residual term resulting from numerical errors and sampling perturbations.

Then, inspired by the entry-wise hard thresholding segmentation thought, we further select out the significant elements from $\mathbf{S}$, and get the third sparse item $\mathbf{St} \in \mathbb{R}^{D\times m}$ as follows:

$$\begin{aligned}&\mathbf{St} = \mathrm{P}_{\Omega}(\mathbf{S}) \\ &\Omega : |\mathbf{S}_{i,j\in\Omega}| \neq 0 \text{ and } \geq \left|\mathbf{S}_{i,j\in\bar{\Omega}}\right|, |\Omega| \leq k\end{aligned} \tag{18}$$

Here, $\mathrm{P}_{\Omega}(\mathbf{S})$ could acquire the non-zero subsets of the first k largest entries of $\mathbf{S}$. k is set by $j * b$.

As mentioned above, the obtained $\mathbf{L}$ has the low-rank feature. $\mathbf{R_Sn}$, $\mathbf{S}$ and $\mathbf{St}$ own sparse characterization. That contributes to realizing an effective anomaly detection. Thus, we construct the anomaly detection detectors as follows:

$$\delta_{\mathbf{B}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{A}}) = (\mathbf{r}_{\mathbf{A}} - \boldsymbol{\mu}_{\mathbf{B}})^{T}\,\mathrm{K}_{\mathbf{B}}^{-1}(\mathbf{r}_{\mathbf{A}} - \boldsymbol{\mu}_{\mathbf{B}}) \tag{19}$$

$$\delta_{\mathbf{B}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{A}}) = \mathbf{r}_{\mathbf{A}}^{T}\mathbf{R}_{\mathbf{B}}^{-1}\mathbf{r}_{\mathbf{A}} \tag{20}$$

Here, $\mathbf{A}$ and $\mathbf{B}$ could be selected from $\{\mathbf{St}/\mathbf{S}/\mathbf{L}+\mathbf{S}/\mathbf{L}+\mathbf{St}/\mathbf{R_Sn}\}$, and $\{\mathbf{L}/\mathbf{S}/\mathbf{St}/\mathbf{R_Sn}\}$, respectively. $\boldsymbol{\mu}_{\mathbf{B}}$, $\mathbf{R}_{\mathbf{B}}$ and $\mathbf{K}_{\mathbf{B}}$ are defined as follows:

$$\boldsymbol{\mu}_{\mathbf{B}} = (1/N)\sum_{i=1}^{N}\mathbf{r}_{\mathbf{A}_i} \tag{21}$$

$$\mathbf{R}_{\mathbf{B}} = (1/N)\sum_{i=1}^{N}\mathbf{r}_{\mathbf{A}_i}\mathbf{r}_{\mathbf{A}_i}^{T} \tag{22}$$

$$\mathbf{K}_{\mathbf{B}} = (1/N)\sum_{i=1}^{N}(\mathbf{r}_{\mathbf{A}_i} - \boldsymbol{\mu}_{\mathbf{B}})(\mathbf{r}_{\mathbf{A}_i} - \boldsymbol{\mu}_{\mathbf{B}})^{T} \tag{23}$$

in which N denotes the total number of the pixels in $\mathbf{A}$. Thus, we define $(\mathbf{A},\mathbf{B})$ with the following five cases: $(\mathbf{S},\mathbf{S})$, $(\mathbf{S},\mathbf{L})$, $(\mathbf{L}+\mathbf{S},\mathbf{L})$, $(\mathbf{L}+\mathbf{St},\mathbf{St})$, and $(\mathbf{R_Sn},\mathbf{R_Sn})$. Correspondingly, the generated ten detectors are $\delta_{\mathbf{S}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{S}})$, $\delta_{\mathbf{L}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{S}})$, $\delta_{\mathbf{L}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{L}+\mathbf{S}})$, $\delta_{\mathbf{St}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{L}+\mathbf{St}})$, $\delta_{\mathbf{R_Sn}}^{\text{RX-AD}}(\mathbf{r}_{\mathbf{R_Sn}})$, $\delta_{\mathbf{S}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{S}})$, $\delta_{\mathbf{L}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{S}})$, $\delta_{\mathbf{L}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{L}+\mathbf{S}})$, $\delta_{\mathbf{St}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{L}+\mathbf{St}})$, and $\delta_{\mathbf{R_Sn}}^{\text{R-AD}}(\mathbf{r}_{\mathbf{R_Sn}})$. Among above ten anomaly detection results, the RSOSP takes all of them through EPF spatial filter processing, thus getting the detection result maps $\mathrm{Amap}_{t_i} = \{d_1, d_2, ..., d_9, d_{10}\}$, respectively.

Then, the RSOSP constructs the feedback set Bmap_{t_i} to extend the original HSI data, of which Bmap_{t_1} is extracted by the partial spectral and spatial information from Amap_{t_i}. Here, $\mathrm{Bmap}_{t_i} = [\,\mathrm{B_dt}(d_1), \mathrm{B_dt}(d_3), d_4, d_5, \mathrm{B_dt}(d_6), d_7, \mathrm{B_dt}(d_8), d_9, d_{10}\,]$. For a given anomaly detection map, let $\mathbf{H}$ denotes the non-zero subsets of the first $cardf$ largest entries of d_w.

$$\begin{aligned}&\mathbf{H} = \mathrm{P}_{\prod_w}(d_w) \\ &\prod_w : |d_{w_{i,j\in\prod_w}}| \neq 0 \text{ and } \geq \left|d_{w_{i,j\in\bar{\prod}_w}}\right|, |\prod_w| \leq cardf\end{aligned} \tag{24}$$

in which projection operator $\mathrm{P}_{\prod_w}(d_w)$ is the projection of a matrix d_w on the set $\prod_w$.

The threshold ϑ_w is adaptively generated by the mean value

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of these selected entries:

$$\vartheta_w = \frac{1}{cardf} \sum_{(i,j)\in \Pi_w} d_w(i,j) \tag{25}$$

Accordingly, the $\mathrm{B_dt}(d_w)$ is obtained by retaining the elements in d_w whose values are less than ϑ_w and setting the remaining elements to zero, defined as followed:

$$\mathrm{B_dt}(d_w) = \begin{cases} d_w(i,j), & d_w(i,j) < \vartheta_w \\ 0, & otherwise. \end{cases} \tag{26}$$

As some new bands, Bmap_{t_i} is used for supplying HSI data $\mathbf{Y} \in \mathbb{R}^{D\times b}$. Thus, the HSI data $\mathbf{Y}^{'} \in \mathbb{R}^{D\times(b+9)}$ is generated. Next, the same operations will be conducted in $\mathbf{Y}^{'}$, until the number of iterations is reached.

B. Algorithm description

The algorithm of the RSOSP is shown in Algorithm 1. The RSOSP first applies SCS to obtain most of image information, then imposes the low-rank constraints to get the background components. Then, RSOSP adopts projection strategy to get sparse components in original HSI data space and compressed sensing band domain. About the obtained various components, we conduct AD and combine the optimization thought to strengthen HSI data, thus realizing a superior detection of anomalies.

Algorithm 1 RSOSP

Input: HSI data $\mathbf{Y} \in \mathbb{R}^{D\times b}$;

1: **Initialize**: m, j, $cardf$, $iter_\max$ = Iterations, and Gaussian random matrix $\phi_m \in \mathbb{R}^{b\times m}$, $i = 1$;

2: **while** $i <= iter_\max$

3: Calculate the low-rank item $\mathbf{L}$ with (11) and (12);

4: Generate three sparse items $\mathbf{R_Sn}$, $\mathbf{S}$ and $\mathbf{St}$ via (14), (16) and (18);

5: Perform anomaly detection process with (19) to (20), and get Amap_{t_i};

6: **If** i surfs the expected condition

7: **then** Take Amap_{t_i} as the final detection set **break**;

8: **else** Expand $\mathbf{Y}$ through Bmap_{t_i};

9: **end**

10: **end while**

Output: The final detection result map Amap_{t_i}

IV. Experiment Analysis

A. Datasets

This paper uses multiple hyperspectral datasets to verify the detection performance of different methods. Fig. 2 shows the false-color and ground truth (GT) map of these datasets, respectively.

1) HYDICE Urban Dataset: The first dataset is the HYDICE Urban scene. It was collected by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor with a spatial resolution of 1 m. This image contains 80×100 pixels with 175 spectral bands. The roofs and cars are regarded as anomalies.

2) Cat Island Dataset: The second dataset used in our research is the Cat Island dataset. The Cat Island scene was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, which has a spatial resolution of 17.2 m. It contains 150×150 pixels and 188 spectral bands. The main anomalies are airplanes.

3) Bay Champagne Dataset: This scene was also captured by the AVIRIS sensor with a 4.4 m spatial resolution. The scene consists of 100×100 pixels and has 188 spectral bands after the removal of unsuitable bands. The artificial reef is regarded as an anomaly target.

4) Pavia Center Dataset: The fourth hyperspectral dataset employed in our study is the Pavia Center dataset. The Pavia Center image was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. There are 100×100 pixels for each band, with each pixel having a spatial resolution of 1.3 m. It includes 102 spectral bands. The anomalies are several vehicles.

5) Texas Coast Dataset: This Texas Coast image was also captured by the AVIRIS sensor. The dataset contains 207 bands and has a spatial size of 100×100 pixels and a spatial resolution of 17.2 m. Buildings are usually seen as anomalies.

6) Salinas Dataset: The final hyperspectral dataset employed in our study is the Salinas dataset. The Salinas scene was captured by the AVIRIS sensor with a spatial resolution of 3.7 m. It covers an area of 512×217 pixels and contains 224 bands. The scene includes 12 block targets embedded in the farmland.

B. Comparison methods and Evaluation criteria

1) Comparison methods: We compare and analyze the detection performance with the following methods: Two statistical theory-based methods, namely RX[31] and PTDPS[76]. RX is the basic method in Hyperspectral anomaly detection. PTDPS is an unsupervised hyperspectral anomaly detection method with a model-free design, integrating point-set topology and information theory. A CR-based method, namely CRD [41], which represents the background pixels via the linear combination of spatial neighbors, thus realizes anomaly detection. A state-of-the-art LRR-based method, namely GTVLRR[77], which incorporates the graph regularization to keep local geometrical structure in the low-rank representation model. Three low-rank and sparse decomposition methods, namely LRSNCR[78], LSMAD [51] and OSPGoDec [62], where LRSNCR improves the robust principal component analysis (RPCA) model to detect anomalies. Two tensor-based methods, namely TLRSR [47] and GNLTR[79], where TLRSR method separates the low-rank background and anomalies via tensor-based low-rank and sparse representation, and GNLTR generates the background and sparse tensor using nonconvex penalty functions and the $L_{2,1}$ norm, respectively. A data decomposition method, namely LEBSR[61], which decomposes the HSI data into the background endmembers and the corresponding abundances, and detects anomalies via the reconstructed background residuals. Four deep learning-based methods, namely GAED[21], PDBSNet [80], DFAN [81] and TGFA[82]. Specifically, GAED constructs AE network to reconstruct HSI and utilizes a guided module to enhance

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background represent; PDBSNet adopts a blind-spot strategy to reconstruct the center pixel using only its neighbors, and breaks the spatial correlation between anomalies and background pixels through pixel-shuffling downsampling; DFAN combines an adaptive aggregation model with a novel loss function to distinguish background pixels and anomalies; and TGFA obtains abundance matrices from HSI, reconstructs them using ASCR-Former, and applies fractional abundance attention to fuse abundance and abundance residual information. The detail parameters of compared methods are based on relevant literature settings. The parameter of the proposed method covers *m*, *j*, *card_f*, and *iterations*, which will be discussed in Section III-D.

2) Evaluation criteria: We display the detection map for a visual observation. Then, the 3-D receiver operating characteristic (ROC) curve of P_d, P_f and τ, and their corresponding 2-D ROC curves of (P_d, P_f), (P_d, τ) and (P_f, τ) are provided to a comprehensive view[83]. For quantitative analysis, the area under curve (AUC) of (P_d, P_f), (P_d, τ) and (P_f, τ) are denoted as $AUC_{(D,F)}$, $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$, respectively. The $AUC_{(D,F)}$ is the primary evaluation metric used to assess the effectiveness of anomaly detectors. The $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ reflect the anomaly target detection ability and background suppression ability, respectively. On the whole, $AUC_{AOD} = AUC_{(D,F)} + AUC_{(D,\tau)} - AUC_{(F,\tau)}$ index represents the overall detection probability of the detector. And $AUC_{SNPR} = AUC_{(D,\tau)} / AUC_{(F,\tau)}$ represents the overall detection performance, signal to noise probability ratio (SNPR), and the separation degree of background and target[84].

In addition, the normalized background anomaly box-plot separation map is applied to evaluate. In this paper, the detection result is firstly normalized to 0-1. The background and anomalies distributed within the [10%, 90%] range is represented by the boxes. Thus, the box-plot reflects the separation degree, target detection ability and suppression ability by the gap between background pixels and anomalies, the length of the anomaly box, and the length of background box, respectively. Last, the running time is considered to evaluate the computational efficiency of various methods.

C. Experimental Results and Analysis

The detection result map obtained by different methods under six datasets are shown in Fig. 2. Table I to Table III show the quantity result values of different methods. Fig. 3 shows the 3D-ROC curve and the 2D-ROC curves of (P_d, P_f), (P_d, τ) and (P_f, τ) in turns. And the box-plots image between background pixels and anomalies is displayed in Fig. 4.

(1) HYDICE Urban scene

As shown in Fig. 2, the detection result maps of LSMAD, GAED, PDBSNet, LEBSR and the proposed method perform significantly better than other methods in the HYDICE Urban scene. For a quantitative analysis, the values of $AUC_{(D,F)}$, $AUC_{(D,\tau)}$ and AUC_{AOD} of the proposed method are higher than that of other methods under HYDICE Urban scene. Although the AUC_{SNPR} and $AUC_{(F,\tau)}$ values of LEBSR are higher than those of the proposed method, and the AD result map of the proposed method has stronger response of anomalies than LEBSR. Similarly, the box-plot image indicates the proposed method achieves the best overall performance in terms of background anomaly separation degree, anomaly detection ability and background suppression ability. This can be inferred from the lengths of anomaly box and background box, and the distance between these two boxes.

(2) Cat Island scene

In comparison to other methods, the proposed method exhibits relatively obvious anomaly response in Fig. 2. In addition, the $AUC_{(D,F)}$ and AUC_{SNPR} values of RSOSP outperform the comparison methods, demonstrating that RSOSP offers superior performance in anomaly detection. Although the proposed method achieves the second-best performance in terms of $AUC_{(D,\tau)}$ and AUC_{AOD} among the comparison methods, it generates a cleaner background than DFAN. From Fig. 4, the box-plot image is consistent with the anomaly detection result maps and quantitative results. Compared with most comparison methods, the proposed method shows a long anomaly target box and a relatively large gap between background and anomaly boxes while maintaining a shorter background box. That demonstrates the proposed method has a better ability to separate anomalies and background pixels.

(3) Bay Champagne scene

From Fig. 2, most of the methods succeed in detecting anomalies, and the proposed method shows the most obvious response to the anomaly target along with a low false alarm rate. According to the background suppression level, PDBSNet and PTDPS perform better than the proposed method, while the values of $AUC_{(D,F)}$ verify the effectiveness of the proposed method is superior to these two methods by Table II. Although CRD, DFAN and TGFA attain a similar level to the proposed method by $AUC_{(D,F)}$ index. The corresponding detection maps show that the anomaly responses of CRD, DFAN and TGFA are weaker than that of the proposed method. According to Fig. 4, the proposed method exhibits the largest gap between background and anomaly target boxes, indicating superior anomaly-background separability.

(4) Pavia center scene

According to Fig. 2, the background pixels of CRD, TLRSR, GNLTR, and DFAN are relatively blurred. Then, the proposed method, LEBSR, PTDPS, and TGFA realize the detection of most anomalies, among which the proposed method better highlights anomaly targets. Besides, as shown in Table II, the values of $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, AUC_{AOD}, and AUC_{SNPR} of the proposed RSOSP are higher than those of TGFA. In addition, the $AUC_{(D,F)}$, $AUC_{(D,\tau)}$ and AUC_{AOD} value of the proposed method achieves the highest values among all the compared methods. Then, the anomaly-background separation degree and target detection ability both indicate the effectiveness of the proposed method by Fig. 4. Thus, we can infer that the proposed method is significantly superior to other compared methods.

(5) Texas Coast scene

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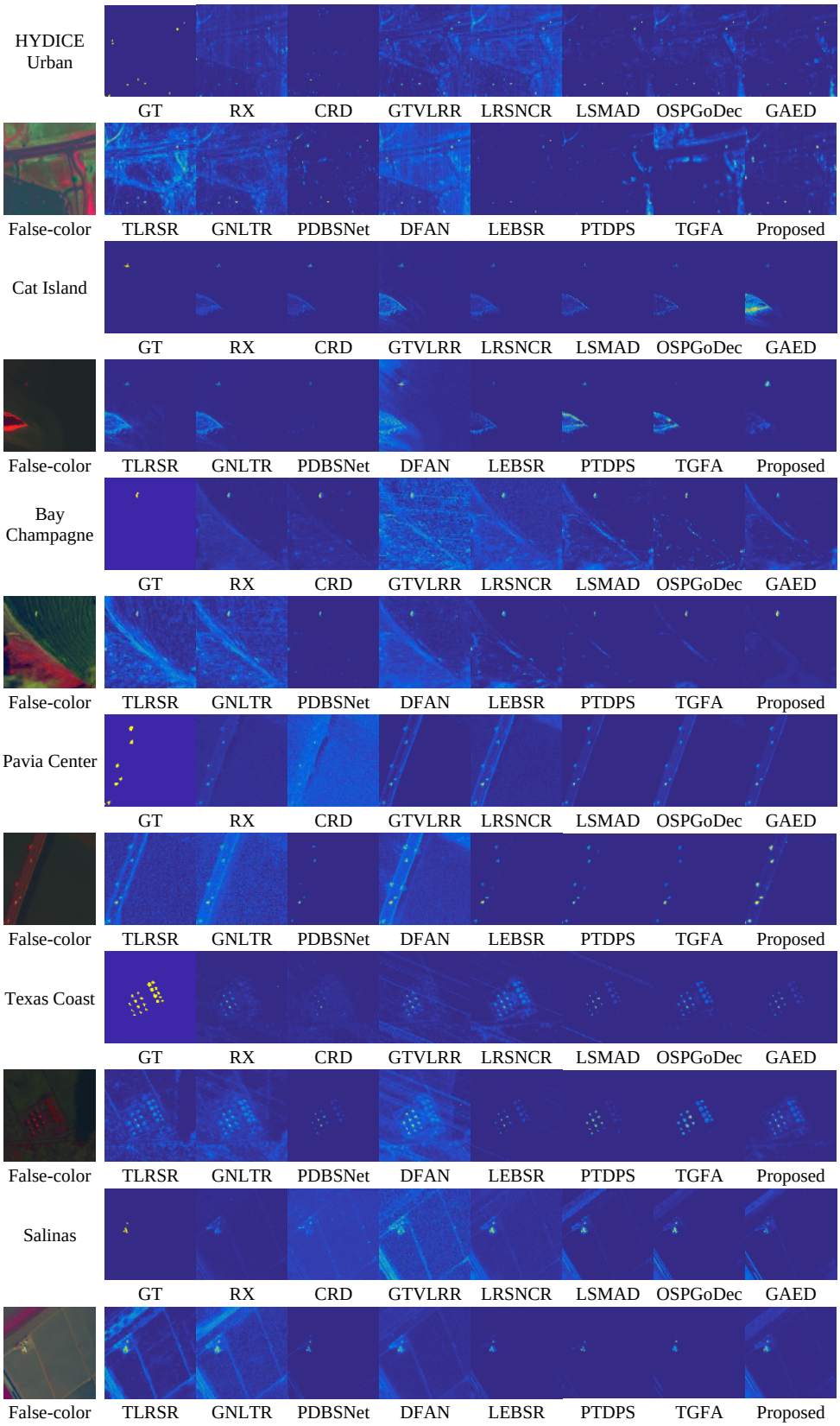


Fig. 2 Anomaly detection map of different methods on six datasets.

RX, CRD, and GTVLRR miss a lot of anomaly pixels in Fig. 2. LRSNCR, TLRSR, GNLTR, and DFAN methods realize anomaly target, with a relatively blurred background. The proposed method, TGFA and OSP-GoDec achieve comparable visual results. Compared with TGFA and OSP-GoDec, the proposed method attains the highest $AUC_{(D,F)}$ value by Table III.

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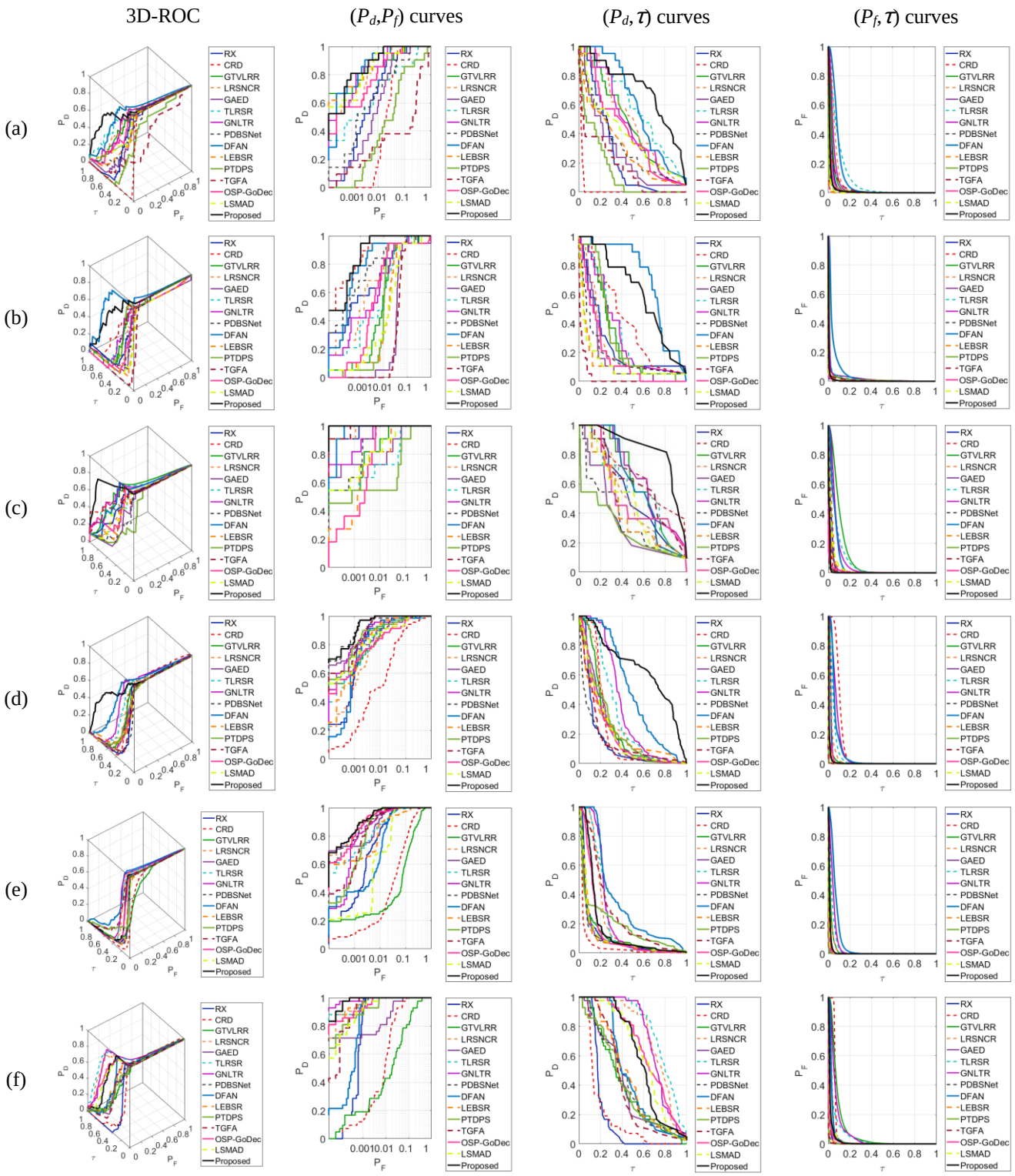


Fig. 3 ROC curves of different methods for six datasets. (From top to bottom) (a) HYDICE Urban, (b) Cat Island, (c)Bay Champagne, (d)Pavia Center, (e)Texas Coast, and (f) Salinas.

Although background suppression level of GAED, PDBSNet and PTDPS are better than the proposed method by the values of $AUC_{(F,\tau)}$ and AUC_{SNPR}, some anomalies are not effectively detected by the detection maps of these three methods. As shown in the box-plot image, the proposed method maintains a balance among anomaly target detection, background suppression and the separation degree of anomalies and background pixels.

(6) Salinas scene

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TABLE I
AUC VALUES OF VARIOUS METHODS ON HYDICE URBAN AND CAT ISLAND DATASETS

Methods	HYDICE URBAN					CAT ISLAND				
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}
RX	0.9862	0.2507	0.036	1.201	6.9633	0.9675	0.2429	0.0075	1.2029	32.494
CRD	0.9528	0.0255	*0.0067*	0.9717	3.831	0.9932	0.3903	0.0086	1.3749	*45.613*
GTVLRR	0.9921	0.4493	0.0391	1.4023	11.501	0.9658	0.2939	0.0213	1.2385	13.805
LRSNCR	0.9957	0.4931	0.0656	1.4232	7.5198	0.9892	0.2176	0.0094	1.1973	23.052
OSP-GoDec	0.9929	0.3993	0.0246	1.3676	16.2503	0.9685	0.1682	*0.0068*	1.1299	24.8902
LSMAD	0.9939	0.3693	0.0118	1.3514	31.1976	0.9542	0.1208	0.0074	1.0677	16.4339
GAED	0.9837	0.2626	0.0155	1.2308	16.95	0.9379	0.1676	0.0205	1.085	8.1764
TLRSR	0.9532	0.5774	0.0946	1.436	6.1024	0.9836	0.3287	0.0197	1.2926	16.684
GNLTR	0.9967	0.465	0.0514	1.4103	9.0409	0.9889	0.3203	0.0161	1.293	19.843
PDBSNet	0.9758	0.3079	0.0118	1.2719	26.152	0.9712	0.1582	**0.0053**	1.124	29.595
DFAN	*0.9978*	*0.595*	0.0905	*1.5023*	6.5771	*0.9943*	**0.7139**	0.0512	**1.657**	13.948
LEBSR	0.9967	0.2817	**0.006**	1.2723	**46.789**	0.9579	0.1182	0.0076	1.0684	15.609
PTDPS	0.9029	0.1363	0.011	1.0282	12.373	0.9808	0.3016	0.0085	1.2738	35.468
TGFA	0.7671	0.1645	0.0307	0.9009	5.3615	0.7114	0.0271	0.0111	0.7274	2.4347
Proposed	**0.9981**	**0.7002**	0.0198	**1.6786**	*35.4246*	**0.9996**	*0.6076*	0.0077	*1.5995*	**78.5257**

TABLE II
AUC VALUES OF VARIOUS METHODS ON BAY CHAMPAGNE AND PAVIA CENTER DATASETS

Methods	BAY CHAMPAGNE					PAVIA CENTER				
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}
RX	0.9997	0.4914	0.0259	1.4651	18.939	0.9911	0.154	0.0258	1.1193	5.9608
CRD	**0.9999**	*0.6395*	0.0179	*1.6216*	35.811	0.958	0.2354	0.1061	1.0873	2.2191
GTVLRR	0.9963	0.6095	0.1058	1.5001	5.7621	0.9948	0.2405	0.0122	1.2231	19.666
LRSNCR	0.9995	0.5741	0.0601	1.5134	9.5469	0.9937	0.2804	0.0323	1.2418	8.6855
OSP-GoDec	0.9972	0.5141	0.011	1.5003	46.7101	0.9898	0.2042	0.0104	1.1836	19.6681
LSMAD	0.9942	0.4495	0.0287	1.415	15.6505	0.9853	0.2065	0.0079	1.184	26.3053
GAED	0.9895	0.3159	0.0181	1.2873	17.485	0.9944	0.2146	0.0081	1.2009	26.57
TLRSR	0.9841	0.5632	0.0756	1.4717	7.4502	0.9888	0.3337	0.0484	1.2741	6.8959
GNLTR	0.9992	0.6386	0.0719	1.5659	8.8827	0.9964	0.3925	0.0747	1.3142	5.2547
PDBSNet	0.9986	0.3305	*0.0061*	1.323	54.289	0.9504	0.1268	*0.005*	1.0722	25.275
DFAN	*0.9998*	0.575	0.0578	1.517	9.94	0.9943	*0.5064*	0.0665	*1.4342*	7.6139
LEBSR	0.9938	0.4341	0.0213	1.4067	20.418	0.9904	0.2227	0.0058	1.2073	38.577
PTDPS	0.9649	0.2153	**0.0046**	1.1756	47.21	0.9951	0.2223	**0.0008**	1.2166	**285.03**
TGFA	*0.9998*	0.5695	0.0076	1.5618	75.436	*0.999*	0.1718	0.0052	1.1656	33.12
Proposed	**0.9999**	**0.8014**	0.0103	**1.7910**	**78.1514**	**0.9995**	**0.6482**	0.0087	**1.6390**	*74.3564*

PDBSNet, LEBSR, PTDPS and TGFA could better detect anomalies and suppress background than other methods in Fig. 2. Even though some background pixels in the proposed method is not well suppressed. The $AUC_{(D,F)}$ value of the proposed method is greater than that of comparative methods. The Fig. 4 displays the separation ability of the background pixels and anomalies. We can see that the proposed method exhibits relatively favorable overall performance over most comparison methods in terms of the separation degree of background pixels and anomalies, as well as the length of anomaly target box and background box.

In addition, the three 2D-ROC curves in Fig. 3 shown in this section are particularly designed to demonstrate the effectiveness, anomaly target detection ability and background suppression ability of the proposed method, respectively. The results are consistent with the qualitative analysis and visual inspection from the detection map and the box-plot image.

D. Parameter Analysis

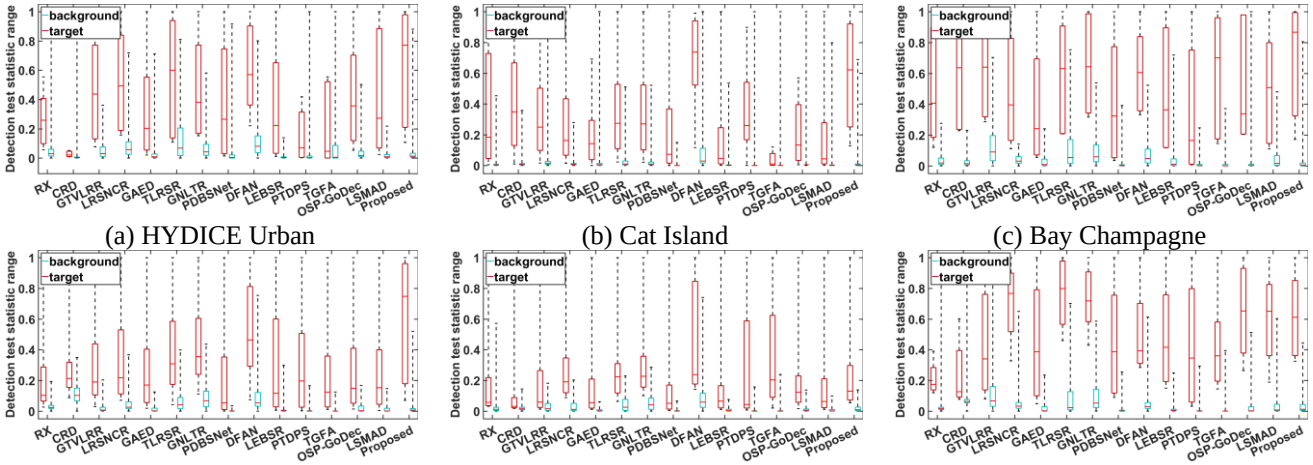
1) *m* and *j*: In this section, we conduct a sensitivity analysis of RSOSP with respect to *m* and *j* while keeping the other parameters fixed. The relationship between the $AUC_{(D,F)}$ score and the values of *m* and *j* across six datasets is shown in Fig. 5. The value of m can be selected from $m_{set}, 2m_{set}, \{2,4,6,8,10,16,24,32\}$, and the value of j can be chosen from $j_{set}, 2j_{set}, \{8,16,24,32,40,80,90\}$, where m_{set} and j_{set} are first initialized by HFC, NWHFC, and MXSVD. For the HYDICE Urban data, the detection result improves as *m* increases and tends to become stable when $6 \le m \le 16$, then

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TABLE III
AUC VALUES OF VARIOUS METHODS ON TEXAS COAST AND SALINAS DATASETS

Methods	TEXAS COAST					SALINAS				
	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{AOD}	AUC_{SNPR}
RX	0.9937	0.1149	0.0136	1.0951	8.481	0.99932	0.1936	0.0175	1.1753	11.031
CRD	0.9015	0.056	0.0166	0.9409	3.3667	0.98533	0.1855	0.065	1.1058	2.8525
GTVLRR	0.8855	0.1216	0.0221	0.985	5.513	0.94773	0.3969	0.0875	1.2571	4.5364
LRSNCR	*0.9988*	0.2218	0.0194	1.2012	11.442	0.99992	*0.7345*	0.0375	*1.6969*	19.597
OSP-GoDec	0.998	0.1504	0.0076	1.1408	19.6961	0.99972	0.66	0.012	1.6597	54.79
LSMAD	0.9862	0.1122	0.007	1.0914	16.007	0.99969	0.6117	0.0177	1.5937	34.5965
GAED	0.9975	0.1029	**0.0031**	1.0973	33.096	0.99517	0.3874	0.013	1.3695	29.709
TLRSR	0.995	0.2307	0.0339	1.1918	6.7994	0.99985	**0.7883**	0.0489	**1.7393**	16.12
GNLTR	0.9979	*0.2536*	0.0451	*1.2064*	5.6222	*0.99993*	0.7279	0.072	1.6558	10.105
PDBSNet	0.9605	0.0914	*0.0051*	1.0468	18.029	0.99981	0.416	0.0066	1.4092	63.234
DFAN	0.9921	**0.3585**	0.0672	**1.2834**	5.3326	0.99953	0.4695	0.0348	1.4342	13.474
LEBSR	0.9885	0.1018	0.0071	1.0832	14.351	0.99987	0.4521	0.0081	1.4439	56
PTDPS	0.9982	0.1853	0.0009	1.1826	**197.57**	0.99951	0.3774	**0.0012**	1.3757	**325.59**
TGFA	0.998	0.2711	0.0064	1.2627	*42.547*	0.99983	0.3926	*0.0057*	1.3868	*69.478*
Proposed	**0.9991**	0.1738	0.0122	1.1607	14.2130	**0.99995**	0.5988	0.0193	1.5795	31.1006



(a) HYDICE Urban (b) Cat Island (c) Bay Champagne

(d) Pavia Center (e) Texas Coast (f) Salinas

Fig. 4 Box plot images of different methods on six datasets.

decreases when $m \geq 22$. The detection score shows no significant changes when $16 \leq j \leq 80$. Thus, the detection performance is sensitive to changes in m under HYDICE Urban scene. The best case is produced when m=10 and j=90. For the Cat Island data, the detection performance improves with increasing m and reaches stable when m is greater than 10. Compared with m, j has a relatively small influence on the detection performance, although a relatively large j may decrease the $AUC_{(D,F)}$ score when m is small. The best performance is achieved when $m = 26$ and j is close to 24. For the Bay Champagne scene, the $AUC_{(D,F)}$ value shows significant growth when $6 \leq m \leq 10$, and then remains at a high level when $16 \leq m \leq 64$. The sensitivity of the detection performance to j gradually decreases as m increases. The proposed method achieves its optimal detection performance with $m = 18$ and $j = 24$. For the Pavia Center image, the $AUC_{(D,F)}$ value increases significantly and reaches its maximum level when $m = 2$ and $j = 80$. Then, the detection result gradually decreases as m increases and remains stable when m exceeds 8. The proposed method is also relatively insensitive to j under Pavia Center dataset. For the Texas Coast dataset, the detection accuracy shows a clear decreasing trend when $2 \leq m \leq 12$, and then gradually recovers as m increases. The sensitivity of $AUC_{(D,F)}$ to j is significantly influenced by m when m takes large values. The optimal detection performance is achieved when $m = 2$ and $j = 24$. For the Salinas dataset, the $AUC_{(D,F)}$ value improves rapidly and remains at a stable level when $4 \leq m \leq 10$. The optimal detection performance is achieved when m and j are set to 10 and 32, respectively. Therefore, the anomaly detection result of RSOSP is sensitive to m on the above six datasets.

2)*Iteration* number and *cardf*: We analyze the sensitivity of the anomaly detection accuracy to the *Iteration* number and *cardf* while keeping other parameters fixed. For parameter analysis, the *Iteration* number is selected from {5,10,20,30}, and *cardf* is selected from {*cardf_pre*, 100,200,1000}, where

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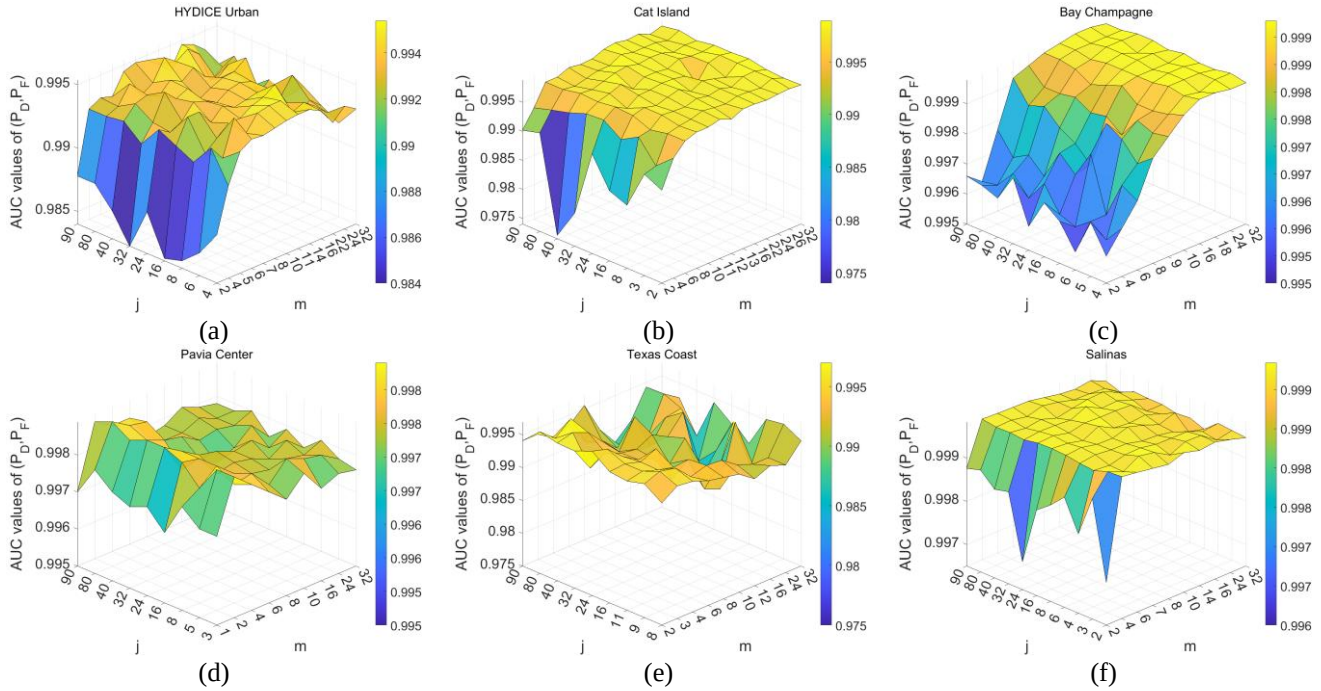


Fig. 5 AUC(P_d,P_f) values of the proposed model under different *m* and *j* settings across six datasets. (a) HYDICE Urban. (b) Cat Island. (c) Bay Champagne. (d) Pavia Center. (e) Texas Coast. (f) Salinas

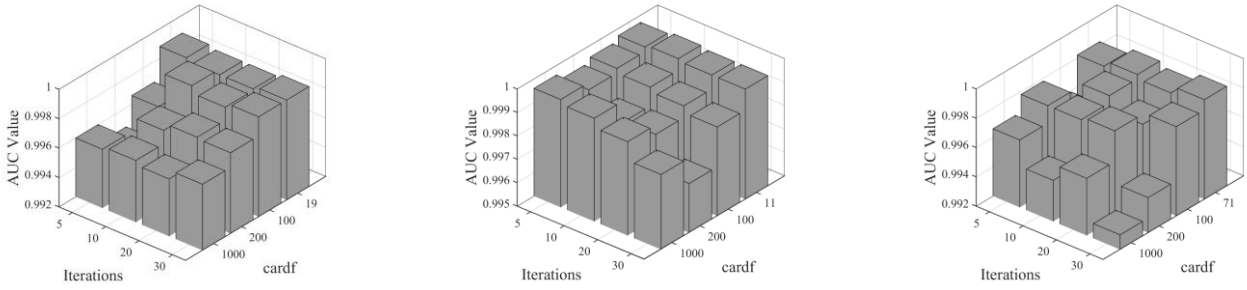


Fig. 6 AUC(P_d,P_f) values of the proposed model under different *card_f* and *Iterations* settings across (a) Cat Island, (b) Bay Champagne, and (c) Pavia Center.

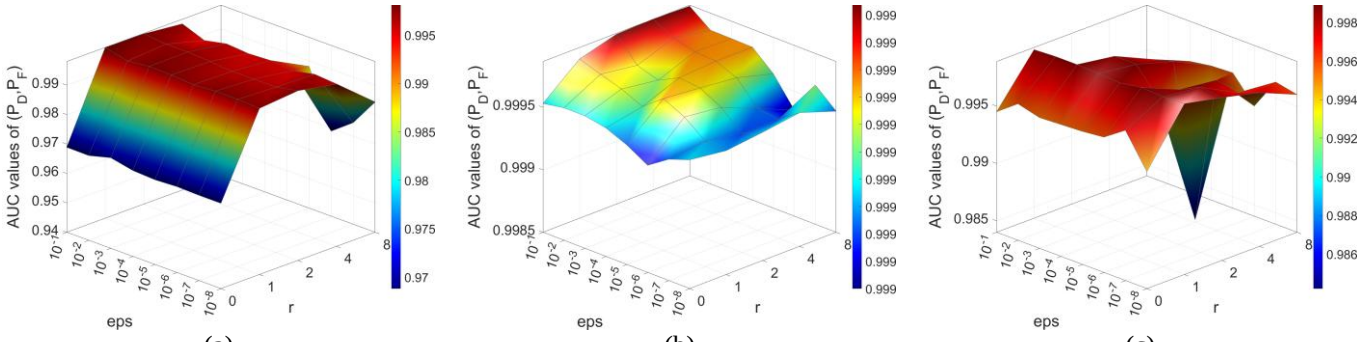


Fig. 7 AUC(P_d,P_f) values of the proposed model under different *eps* and *r* settings across (a) Cat Island, (b) Bay Champagne, and (c) Pavia Center.

cardf_pre is a predefined value estimated according to the expected size of anomaly targets in each dataset. As shown in Fig. 6, for the Cat Island and Pavia Center datasets, the detection accuracy is sensitive to *cardf* and tends to increase as *cardf* decreases. RSOSP achieves optimal detection performance with appropriate values of the *Iteration* number and *cardf*. For the Cat Island, Bay Champagne, and Pavia Center datasets, the optimal AUC values are achieved when *cardf* is set to 19, 11, and 71, respectively. For these three cases, the AUC values reach a high level with a small number

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TABLE IV
AUC(D,F) VALUE OF VARIOUS DETECTORS ON (I) HYDICE URBAN, (II) CAT ISLAND, (III)BAY CHAMPAGNE, (IV)PAVIA CENTER, (V)TEXAS COAST, AND (VI) SALINAS

	$\delta_{S}^{RX\text{-}AD}(\mathbf{r}_{S})$	$\delta_{L}^{RX\text{-}AD}(\mathbf{r}_{S})$	$\delta_{L}^{RX\text{-}AD}(\mathbf{r}_{L+S})$	$\delta_{St}^{RX\text{-}AD}(\mathbf{r}_{L+St})$	$\delta_{R_Sn}^{RX\text{-}AD}(\mathbf{r}_{R_Sn})$	$\delta_{S}^{R\text{-}AD}(\mathbf{r}_{S})$	$\delta_{L}^{R\text{-}AD}(\mathbf{r}_{S})$	$\delta_{L}^{R\text{-}AD}(\mathbf{r}_{L+S})$	$\delta_{St}^{R\text{-}AD}(\mathbf{r}_{L+St})$	$\delta_{R_Sn}^{R\text{-}AD}(\mathbf{r}_{R_Sn})$
(I)	0.9253	0.9116	0.9943	0.9225	**0.9981**	0.9253	0.9462	0.9924	0.9225	**0.9981**
(II)	0.9467	0.3560	0.9987	0.9463	0.9958	0.9467	**0.9996**	0.9987	0.9463	0.9958
(III)	0.9994	0.0090	0.9997	0.9254	**0.9999**	0.9994	0.9993	0.9997	0.9254	**0.9999**
(IV)	0.8786	0.1000	0.9795	0.9917	**0.9995**	0.8858	0.9950	0.9916	0.9915	**0.9995**
(V)	0.6580	0.5300	0.9889	**0.9991**	0.9956	0.6548	0.8154	0.9886	**0.9991**	0.9956
(VI)	0.8902	0.9872	**0.9999**	0.9962	0.9989	0.8788	0.9915	**0.9999**	0.9962	0.9989

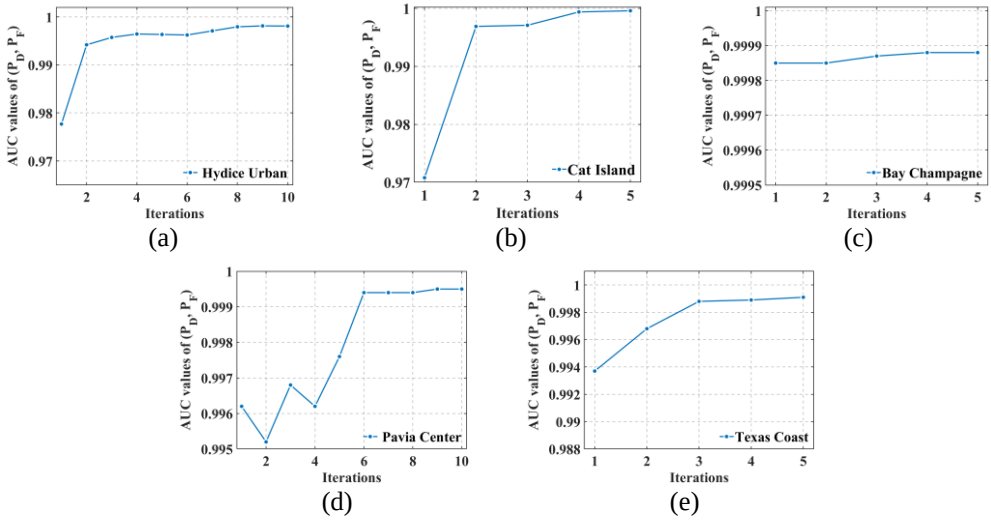


Fig. 8 AUC(P_D,P_F) changes curves respect to the iteration number (a) HYDICE Urban, (b) Cat Island, (c)Bay Champagne, (d)Pavia Center, and (e)Texas Coast

of iterations and then remain relatively stable as the iteration number increases. The optimal *Iteration* numbers are 5, 5, and 10, respectively.

3) *eps* and *r*: We analyze the sensitivity of the detection performance to the blur degree *eps* and filter size r while keeping the other parameters fixed. The parameters *eps* and r are chosen from the sets {1e-1,1e-2, 1e-3,1e-4,1e-5,1e-6,1e-7, 1e-8} and {0,1,2,4,8}, respectively. Fig. 7 illustrates the effects of *eps* and r on the $AUC_{(D,F)}$ scores on three datasets. For the Cat Island dataset, the $AUC_{(D,F)}$ value increases as r increases and tends to become stable when $1 \le r \le 2$. Similarly, the detection performance on the Pavia Center dataset is better when $r = 1$. These results indicate that the $AUC_{(D,F)}$ values on the Cat Island and Pavia Center datasets are not sensitive to *eps*. For the Bay Champagne scene, the $AUC_{(D,F)}$ is relatively high in the upper left corner of Fig. 7, which reaches its maximum where $eps = 1e-1$ and $r = 4$. In general, the anomaly detection result is better when *eps* is equal to 0.1. The detection accuracy can attain a satisfactory level when r is set to 1 or 8.

Based on the above sensitivity analysis, the recommended parameter settings are summarized as follows. First, the values of m can be directly obtained using HFC or NWHFC, and the values of j can be determined by MXSVD. The default value of *Iteration* number is set to 10 because the detection performance becomes stable after several iterations. The value of *cardf* can be set to a small value, e.g., around 10 when anomaly targets are expected to be small. If the anomaly region is roughly estimated to be relatively large, *cardf* can be set to 100. Last, $eps = 0.1$ can be select as the default value. And r can be selected as either 1 or 8.

E. Discussion

1) Multiple detectors analysis

In this paper, we involve multiple detectors via the generated decomposed component, thus conducting the following AD processing. The proposed algorithm stops the feedback and obtains multiple anomaly detection results when the algorithm attains the iteration condition. Table IV lists the $AUC_{(D,F)}$ value of $\delta_{S}^{RX\text{-}AD}(\mathbf{r}_{S})$, $\delta_{S}^{R\text{-}AD}(\mathbf{r}_{S})$, $\delta_{L}^{RX\text{-}AD}(\mathbf{r}_{S})$, $\delta_{L}^{R\text{-}AD}(\mathbf{r}_{S})$, $\delta_{L}^{RX\text{-}AD}(\mathbf{r}_{L+S})$, $\delta_{L}^{R\text{-}AD}(\mathbf{r}_{L+S})$, $\delta_{St}^{RX\text{-}AD}(\mathbf{r}_{L+St})$, $\delta_{St}^{R\text{-}AD}(\mathbf{r}_{L+St})$, $\delta_{R_Sn}^{RX\text{-}AD}(\mathbf{r}_{R_Sn})$ and $\delta_{R_Sn}^{R\text{-}AD}(\mathbf{r}_{R_Sn})$ of the RSOSP under six datasets. From Table IV, RSOSP($\delta_{R_Sn}^{RX/R\text{-}AD}(\mathbf{r}_{R_Sn})$) could achieve the best anomaly detection performance under HYDICE Urban, Bay Champagne, and Pavia Center scenes. As for the Cat Island scene, the best performance was produced by RSOSP($\delta_{L}^{R\text{-}AD}(\mathbf{r}_{S})$), which has obvious advantages over other detectors. About the Texas Coast data, the $AUC_{(D,F)}$ value of RSOSP($\delta_{St}^{RX/R\text{-}AD}(\mathbf{r}_{L+St})$) is higher than that of other detectors. Last, RSOSP($\delta_{L}^{RX/R\text{-}AD}(\mathbf{r}_{L+S})$) shows the best qualitative result for the Salinas data.

2)Convergence analysis

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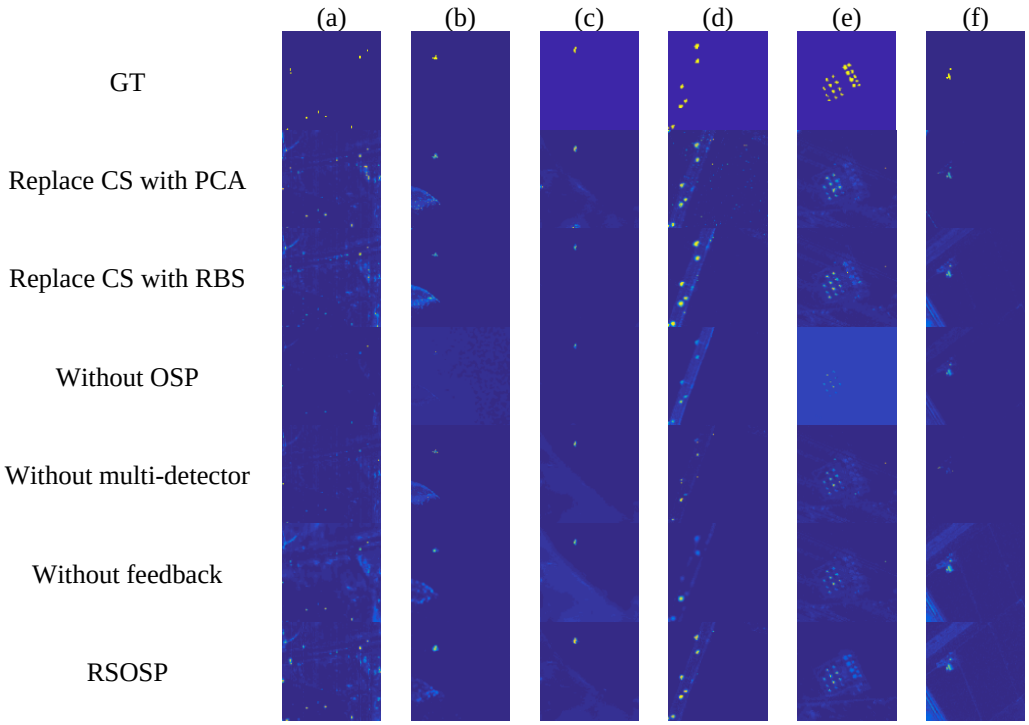


Fig. 9 Detection result maps of ablation on (a) HYDICE Urban, (b)Cat Island, (c)Bay Champagne, (d)Pavia Center, (e)Texas Coast, and (f) Salinas.

TABLE V
AUC(D,F) SCORES FOR ABLATION STUDY ON SIX DATASETS

	HYDICE Urban	Cat Island	Bay Champagne	Pavia Center	Texas Coast	Salinas
Replace CS with PCA	*0.9944*	*0.9987*	0.9996	0.9984	0.9906	0.99948
Replace CS with RBS	0.9906	0.9957	*0.9998*	*0.9990*	*0.9964*	*0.99993*
Without OSP	0.9619	0.9661	0.9997	0.9937	0.7398	0.99991
Without multi-detectors	0.9746	0.9698	*0.9998*	0.9836	0.9931	0.99908
Without feedback	0.9884	0.9922	*0.9998*	0.9959	0.9936	0.99987
RSOSP	**0.9981**	**0.9996**	**0.9999**	**0.9995**	**0.9991**	**0.99995**

Next, the convergence of the proposed method across various datasets is analyzed in Fig. 8, which shows the $AUC_{(D,F)}$ score versus the number of iterations. As the number of iterations increase, the value of $AUC_{(D,F)}$ generally improves and stabilizes in most scenes in Fig. 8. The feedback can provide additional discriminative information and may improve the detection result when the current result still has room for improvement and the feedback image contain effective background response or anomaly response. For Pavia Center dataset, the $AUC_{(D,F)}$ score decreases first and then keeps increasing and stabilizes. This is because the iterative process may introduce estimation errors when the feedback image is noisy or unreliable, leading to temporary performance degradation. In general, the additional improvement provided by the feedback mechanism is limited when the current detection result is already satisfactory. This indicates that the proposed method shows good convergence for the HAD task.

3) Ablation study

To illustrate the necessity of each component, *i.e.*, compressed sampling, OSP, multi-detector and iterative feedback, we comprehensively evaluate the detection results of RSOSP under the following scenarios. 1)without compressed sampling: RSOSP replaces compressed sampling with PCA or random band sampling (RBS) to obtain the low-rank item and then continues the subsequent steps, i.e., OSP, multi-detector and iterative feedback (denoted as "Replace CS by PCA/RBS"); 2) without OSP: only compressed sampling is retained, and the low-rank component is tested with the RX detector (denoted as "Without OSP"); 3) without multi-detector: compressed sampling and OSP are retained, and detection is performed on the sparse component R_Sn, which is obtained by (14), using the RX detector (denoted as "Without multi-detector"); 4) without iterative feedback mechanism: compressed sampling, OSP, and multi-detector are retained (denoted as "Without feedback"). As shown in Fig. 9 and Table V, RSOSP is superior to RSOSP without compressed sampling, OSP, multi-detector, or feedback. It effectively suppresses background pixels, detects anomalies and achieves higher anomaly detection scores than the other cases under six datasets. Thus, the above detection results demonstrate the effectiveness of each component.

4) *Computational complexity*

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TABLE VI
EXECUTION TIME IN SECONDS (RANKING) OF VARIOUS METHODS ON SIX DATA SETS

	HYDICE Urban	Cat Island	Bay Champagne	Pavia Center	Texas Coast	Salinas	Ave
RX	**0.06(1)**	**0.12(1)**	**0.06(1)**	**0.04(1)**	**0.08(1)**	**0.15(1)**	**0.09(1)**
CRD	53.4(12)	76.29(11)	34.27(12)	54.05(12)	43.81(13)	116.8(9)	63.09(9)
GTVLRR	158.7(15)	336.6(15)	194.3(15)	92.65(14)	172.5(15)	302.7(13)	209.6(15)
LRSNCR	11.59(8)	14.69(6)	20.60(9)	12.52(8)	21.05(10)	73.37(8)	25.64(8)
OSP-GoDec	22.68(11)	126.09(12)	34.09(11)	24.29(11)	35.82(11)	256.05(12)	125.85(13)
LSMAD	5.37(6)	14.28(3)	6.82(5)	3.35(5)	6.9(5)	23.44(6)	10.03(5)
GAED	71.49(13)	236.97(14)	32.85(10)	20.34(10)	41.09(12)	213.99(10)	102.8(12)
TLRSR	*2.18(2)*	14.53(4)	*1.85(2)*	2.48(3)	*2.72(2)*	9.52(3)	5.55(3)
GNLTR	3.65(4)	14.61(5)	14.99(8)	2.99(4)	9.52(7)	12.83(4)	9.77(4)
PDBSNet	122.1(14)	196.3(13)	132.6(14)	130.3(15)	131.3(14)	244.1(11)	159.4(14)
DFAN	18.09 (10)	18.09(7)	38.5(13)	65.41(13)	18.09(9)	370.8(14)	88.16(11)
LEBSR	2.28(3)	*3.99(2)*	2.92(3)	*2.20(2)*	2.84(3)	8.74(2)	*3.83(2)*
PTDPS	8.66(7)	30.76(8)	11.38(7)	7.85(7)	12.96(8)	50.15(7)	20.29(7)
TGFA	3.91(5)	53.03(9)	3.31(4)	3.69(6)	3.68(4)	449.7(15)	86.23(10)
Proposed	13.67(9)	22.51(10)	7.79(6)	13.14(9)	7.36(6)	15.86(5)	13.39(6)

Last, we assess the computational complexity by deriving the theoretical complexity and displaying the execution time of the algorithm. The total computational cost of RSOSP is $O\left(pD^2 \bullet \max\left(m,b\right)\right)$, in which p denotes the number of iterations. As shown in Table VI, GTVLRR, PDBSNet, GAED and OSP-GoDec require longer execution time than other methods. Compared with CRD, DFAN, TGFA, LRSNCR and PTDPS, the proposed method exhibits a more efficient average execution time. Then, RX has the shortest running time, followed by TLRSR and LEBSR. In general, the execution time of the proposed method is at a medium level in all the comparison algorithms.

V. CONCLUSION

To exploit the low-rank and sparse prior of HSI data, this paper proposed a space decomposition technology and anomaly detection for HAD. Initially, compressive sampling yields a low-rank component containing most of the dominant information of HSI, and the low-rank constraint further realizes an approximated representation of background. That also saves the computational time of RSOSP, due to the fact that compressive technology reduces the data size of the original data. Besides, the sparse components extracted from the original data space and compressed sensing band domain through Dual-OSP, together with the effective anomaly space projection strategy, preserve complementary and discriminative information for subsequent anomaly detection. In addition, Multiple extraction of background pixels and anomalies information in the detection map feed and strengthens the original data, thus realizing a more outstanding AD. Last, experiments on multiple real hyperspectral datasets verify that the proposed RSOSP obtains a superior detection result than that of other compared methods.

As mentioned above, the detection result is sensitive to the compressed HSI data, and the compressed sampling result is related to the sampling matrix. In the future study, we will discuss the applicability of other sampling matrices to our method. In addition, we may discuss the reasonableness of the application of the obtained low-rank and sparse components in other fields.

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